Machine Learning vs. Deep Learning: Customer Churn Prediction and Machine Translation

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Machine learning (ML) is a subfield of artificial intelligence in which algorithms learn patterns from data, while deep learning (DL) is a subset of ML that uses multilayer neural networks to learn representations. Traditional ML often relies on analyst-selected variables, whereas DL can learn more complex representations from data. Although both approaches can use labeled or unlabeled data, their suitability depends on the problem, available data, computational requirements, and expected performance. The two examples here are: customer churn prediction with a support vector machine (SVM), often considered well-served by traditional ML, and machine translation with Transformer-based neural networks, a form of DL.

Machine Learning: Customer Churn Prediction

Real-World Application

Churn prediction identifies customers at risk of discontinuing a service so retention efforts can be targeted. An SVM is a supervised algorithm that classifies data by finding the boundary that best separates classes (IBM, 2023). Gordini and Veglio (2017) applied a tuned SVM to churn in B2B e-commerce, benchmarking it against logistic regression, a neural network, and a classic SVM. It performed best on noisy, imbalanced, nonlinear marketing data. Roberts et al. (2022) used subscription behavior from around 10,000 customers of a major academic publisher to estimate churn through five conventional ML models, including an SVM.

Why Machine Learning?

Churn data are structured: each customer is a row of defined attributes, and analysts can select variables to include directly, the manual step associated with traditional ML. In Roberts et al. (2022), random forest and gradient boosting achieved the strongest results, with random forest

achieving 90% classification accuracy up to 200 days ahead. Across 45 tabular datasets, Grinsztajn et al. (2022) found that tree-based models outperformed neural network models evaluated on many medium-sized datasets (around 10,000 samples), even without accounting for faster training. SVM is therefore best read as one representative of a family of traditional methods, and a tree ensemble may be the stronger choice in practice.

Why NOT Deep Learning?

Although DL can be applied to structured data, its advantages over conventional ML have not consistently been demonstrated on typical tabular churn datasets. Learning variables from raw, high-dimensional input adds little when there are already table columns. Grinsztajn et al. (2022) identified where neural networks struggle on tabular data: robustness to uninformative variables, preserving data orientation, and learning irregular patterns. A tuned SVM outperformed a neural network in Gordini and Veglio (2017), and Roberts et al. (2022) reached high accuracy without DL. Deep networks would add training cost, tuning effort, and opacity without demonstrated gain, although larger, more complex datasets change that calculation.

Deep Learning: Machine Translation

Real-World Application

Machine translation converts text between languages. Google Research reported replacing the original system in Google Translate with models that pair a Transformer encoder with a recurrent decoder, which improved quality by an average of 5 BLEU points across more than 100 languages (Caswell & Liang, 2020). Meta's No Language Left Behind (NLLB-200) system supports translation across 200 languages and reported an average of 44% improvement in BLEU-based translation quality compared with previous state-of-the-art systems evaluated in the study (NLLB Team, 2024).

Why Deep Learning?

Translation maps one sequence of words into another, and meaning depends on context throughout the sentence. Vaswani et al. (2017) introduced the Transformer, which relies entirely on attention mechanisms instead of recurrence to model relationships within sequences. Trained on 4.5 million English-German sentence pairs, it outperformed the state-of-the-art WMT 2014 English-to-German benchmark by more than two BLEU points, and its English-to-French model trained in 3.5 days on eight GPUs, a fraction of the training cost of earlier leading models. Attention alone can capture relationships between distant words (Vaswani et al., 2017).

Why NOT Machine Learning?

Conventional ML classifiers are well-suited to predicting outcomes from predefined variables, such as spam or churn. Machine translation differs because it requires generating variable-length sequences while preserving context. The output is an open-ended sentence rather than a label, and count-based variables would eliminate the context of word order and long-range relationships that carry most of the sentence's meaning; the NLLB pipeline shows where each approach belongs. For the narrow task of identifying a text language, the team used fastText, a lightweight linear classifier that prioritizes simplicity, speed, and explainability, while translation itself relied on large Transformer models (NLLB Team, 2024). Earlier phrase-based systems could translate, but Google reported that moving to neural translation in 2016 produced results comparable in size to the later five-BLEU improvements (Caswell & Liang, 2020).

Comparison, Conclusion, and Implications

Comparison

The examples differ in data structure and output form. Churn data are structured rows that map to a yes-no churn outcome or estimated churn risk, allowing analysts to define variables

and evaluate predictive performance using the appropriate validation methods. Translation, on the other hand, involves unstructured input sequences that must be transformed into new sequences in the target language, so variables must be learned from large-scale data. These differences can be directly compared: manual versus automatic feature extraction, modest versus large data and computing needs, and simpler versus more complex models.

Implications

These examples demonstrate that model selection should be guided by the problem's characteristics rather than algorithmic complexity. For structured data like customer records, conventional ML methods such as SVMs, random forests, and gradient boosting provide practical baselines. Deep learning may be justified when additional data complexity produces measurable performance improvements, but these benefits must be evaluated against computational costs and interpretability requirements. Both applications require careful validation: churn models must account for class imbalance (Gordini & Veglio, 2017) and evolving customer behavior, while translation systems must be evaluated for linguistic accuracy, cultural context, and cross-lingual performance.

Conclusion

Traditional machine learning and deep learning offer different advantages depending on the application. Customer churn prediction demonstrates how conventional ML can provide accurate, actionable results from structured data without requiring more complex neural architectures. Machine translation illustrates the advantages of deep learning for problems involving contextual relationships and sequence generation. Neither approach is universally superior. Effective model selection requires balancing predictive performance, data characteristics, computational efficiency, and the practical requirements of the application.

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AI Disclosure

Generative AI was used to help identify sources, both case studies and supporting references, based on the thesis and main proposition. It was also used to revise the draft, identify logical inconsistencies, and format references in APA style. All content was cross-checked and reviewed before submission.